

ALI REZAALI

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RESEARCH INTERESTS

Remote Sensing & Earth Observation • GeoAI & Explainable Machine Learning • Climate & Urban Hazard-Risk Assessment • Spatiotemporal Modeling

EDUCATION

M.Sc. in Geospatial Information Systems (GIS)

Sep 2024 – Expected Oct 2026

K. N. Toosi University of Technology, Tehran, Iran

CGPA: 4/4 (19.44/20)¹ | Rank: 1st of 32

B.Sc. in Geomatics Engineering

2020 – 2024

K. N. Toosi University of Technology, Tehran, Iran

CGPA: 3.64/4 (17.51/20) | Rank: 3rd of 42

PEER-REVIEWED PUBLICATIONS

- (1) **Rezaali, A., & Pilehforooshha, P.** (2026). "Impact of the 2025 California Wildfires on Mountain Lion (*Puma concolor*) Habitat: A Remote Sensing and Species Distribution Modeling Framework." ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, X-4/W8-2025, 609–616. doi.org/10.5194/isprs-annals-X-4-W8-2025-609-2026. • **Oral Presentation**, 8th ISPRS Geospatial Conference, Tehran, Iran, Dec 2025. • **Related seminar**: "A Review of Key Factors and Methods for Spatial Species Distribution Modeling," M.Sc. Seminar, K. N. Toosi University of Technology, Sep 2025 (in Persian).
- (2) **Rezaali, A., Ebadi, H., Farhadi, H.** (2025). "Monitoring Spatiotemporal Changes of Water Bodies Using Remote Sensing Data in Google Earth Engine (Case Study: Lake Urmia)." Journal of Geomatics Science and Technology (JGST). 2025; 14 (3): 6. doi.org/10.61186/jgst.14.3.69 • Double-Blind Peer-Reviewed; In Persian; English abstract available online.
- (3) **Rezaali, A., Ebadi, H., Farhadi, H.** (2025). "Detection and Monitoring of Water Body Changes Using Innovative Thresholding of Spectral Indices (Case Study: Gorgan Gulf and Miankaleh Wetland)." Scientific-Research Quarterly of Geographical Data (SEPEHR), 34(134), 33-60. doi.org/10.22131/sepehr.2024.2041037.3096 • Double-Blind Peer-Reviewed; In Persian; English extended abstract available online.

Manuscripts in Preparation

- (1) **Rezaali, A., & Pilehforooshha, P.** "Modeling Urban Heat Hazard using an Explainable Spatiotemporal GeoAI Fusion Framework Across Climatic Archetypes." In Preparation for a Q1 Remote Sensing/Geospatial Journal.
- (2) **Rezaali, A., & Pilehforooshha, P.** "IPCC-Based High-Resolution Spatiotemporal Urban Heat Risk Modeling." In Preparation for a Q1 Remote Sensing/Geospatial Journal.

THESES

(1) Master's Thesis

Expected Oct 2026

- **Supervisor:** Dr. Parastoo Pilehforooshha | K. N. Toosi University of Technology
- **Title:** Spatiotemporal Modeling of Urban Heat Risk with Explainable GeoAI
- Developing a multi-pillar, explainable GeoAI framework for spatiotemporal surface urban heat island (SUHI) and urban heat risk (UHR) modeling – integrating multi-sensor Earth observation with exposure and vulnerability data within an IPCC hazard-exposure-vulnerability risk framework – to model and explain dynamic urban climate hazards across multi-season, multi-year time-steps and two contrasting archetypes (Tehran & Singapore), integrating 10m-resolution Earth observation, downscaled ERA5-Land reanalysis, 10m LST downscaling, and 3D Local Climate Zone (LCZ) modeling. Designed for transferability and global scalability.

(2) Bachelor's Thesis

Defended Sep 2024, Grade: 20/20

- **Supervisor:** Dr. Hamid Ebadi | **Co-supervisor:** Dr. Hadi Farhadi | K. N. Toosi University of Technology
- **Title:** Time Series Monitoring of the Surface Water Body Changes Using Remote Sensing Satellite Imagery in Google Earth Engine (Case Study: Iran)
- Nationwide, pixel-based time-series analysis of Iran's surface water dynamics (2018–2024, seasonally) using Sentinel-2 (10m) and Landsat-8 (30m) in GEE, with focused case studies on four water body types — the Karun River, Lake Urmia, Gorgan Gulf, and Dez Dam. Comparatively evaluated 8 spectral indices, quantified seasonal and spatial trends, and assessed the impact of topographic correction on extraction accuracy (up to 0.95 OA). doi.org/10.5281/zenodo.22727244

¹ Coursework completed, thesis grade pending.

HONORS & AWARDS

Exceptional Talent Ph.D. Admission – Eligible, Not Pursued • K. N. Toosi University of Technology • Guaranteed by #1 in class rank in the M.Sc. program; pursuing international PhD study instead.	Expected 2026
Exceptional Talent Admission – M.Sc. GIS • K. N. Toosi University of Technology • Direct admission to the Master’s program, reserved for top-ranked undergraduate students.	Aug 2024
Dean’s List (2023–24) • K. N. Toosi University of Technology • Issued by Dr. Yazdan Amerian (Associate Professor of Geodesy).	July 2024
Top 5 Entry, National M.Sc. Entrance Exam • Geomatics Engineering, Iran – ranked 5th nationally, among nearly 2,000 applicants.	July 2024
Top 1% Iranian National University Entrance Exam (<i>Konkoor</i>) • Secured a high rank in Iran’s highly competitive national university entrance exam.	July 2020

SELECTED TECHNICAL PROJECTS

- **Spatiotemporal WebGIS Dashboard** (M.Sc. Coursework): Full-stack interactive dashboard on PostgreSQL/PostGIS for exploring spatiotemporal urban incidence data, with dynamic time filtering and hotspot visualization.
- **Fuzzy Habitat Suitability Modeling** (M.Sc. Coursework): Mamdani Fuzzy Inference System (MFIS) applied across California.
- **LiDAR & 3D Point Cloud Analysis** (B.Sc. Coursework): Unsupervised and supervised classification (K-Means, BIRCH, Random Forest, SVM) with engineered geometric features (PCA, inclination angle) improving overall accuracy from 0.82 to 0.92.
- **Digital Image Processing & Change Detection**: Python-based segmentation and thresholding for change detection (OpenCV).

PROFESSIONAL EXPERIENCE

GIS & Photogrammetry Intern – Iran Naghshine • Collaborated on digital mapping and orthomosaic preparation for water transmission line projects. • Produced GIS maps for cities/counties across Iran using AgiSoft Metashape and Global Mapper.	Jun–Aug 2024
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TECHNICAL SKILLS

Programming	Python, JavaScript (GEE), R (dismo, spThin), SQL
Geospatial/ML Python Libraries	Earth Engine, Xarray, XEE, rioxarray, GeoPandas, geemap, Rasterio, Shapely, OSMnx, NumPy, Pandas, SciPy, Matplotlib, OpenCV, Scikit-Learn, Scikit-Image, TensorFlow, SHAP, XGBoost, LasPy, RPy2, Open3D
WebGIS & Databases	PostgreSQL/PostGIS, HTML, CSS, JS, Leaflet, OpenLayers, GeoServer, Django
GIS, Remote Sensing & Photogrammetry Tools	ArcGIS Pro, QGIS, ENVI, AgiSoft Metashape, Global Mapper, AutoCAD, Civil3D
Geospatial Data Sources & Platforms	Sentinel-1/2, Landsat 8/9, VIIRS (NTL/LST), ERA5/-Land, WorldClim/Bioclim, MODIS, CHIRPS, LiDAR, OSM, WorldPop, GBIF, GHCN-h, AW3D30, FABDEM
Design, Visualization & Cartography	Adobe Photoshop, Adobe Illustrator, CorelDRAW, Blender
Developer Tools	Google Colab, GitHub, Jupyter Notebook, VS Code, PyCharm

CERTIFICATIONS

Esri MOOCs: Cartography (2024); GIS for Climate Action (2024); Spatial Data Science (2024); Modern Geo Apps (2024); Going Places with Spatial Analysis (2025); The ArcGIS Imagery MOOC: Foundations and Frontiers (2025).

LANGUAGES

English (Proficient) • Persian (Native, Fluent) • French (Elementary)

REFERENCES

Available upon request.